

Analysis Of Cyber Threat Detection And Emulation Using MITRE Attack Framework

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Abstract— With a rapid increase in Cyber-attacks, Threat hunters such as Cyber Threat Intelligence (CTI) and their teams requires to analyze different techniques being employed by adversaries to hit a target objective. The attacker objectives can be from entering in your network, accessing system files and folders remotely, getting higher system privileges, stealing confidential passwords etc. to destroying systems and network. Pre attack patterns defined in enterprise knowledge base can play a major role to track adversary techniques and procedures in order to defend and response from such attacks. Anomalous and intrusion activities need to be unfolded by the approach adversaries are adopting to demolish secure enterprise networks. An appropriate system is required to better handle modern attack approaches and strategies used by attackers in order to identify vulnerabilities and successfully defend network channels. In this paper, we present an in-depth analysis of different threat detection methods and how to mitigate their impacts using MITRE ATT&CK framework. This framework is an extensively and freely accessible knowledge repository of tactics, techniques and procedures (TTPs) to gain an insight into what techniques adversaries are using in real time applications which aids in developing robust threat controlling programs in private sector, government, and in cybersecurity community.

Keywords— Threat Intelligence, Cyber-attacks, Network Security, vulnerability.

I. INTRODUCTION

The exponential growth of the Internet interconnections has led to a significant growth of cyber-attack incidents often with disastrous and grievous consequences. People are sharing personal and confidential information to one another by means of networks. At the same time, with the rapid use of Internet in day-to-day life, intrusion activities are increasing at a great pace [1]. Malware is the primary choice of weapon to carry out malicious intents in the cyberspace, either by exploitation into existing vulnerabilities or utilization of unique characteristics of emerging technologies [2]. The development of more innovative and effective malware defense mechanisms has been regarded as an urgent requirement in the cybersecurity community. Information technology, Digital Services using API, Artificial Intelligence, Data Science, Mobile banking, Automation spawn tremendous contribution in the direction of growing the efficiency of most business application processes [3]. Recent advances in data intensive information processing systems is becoming increasingly important to make decisions in business organizations. Various kinds of analytical measures were employed and plotted them against the different data sets on a graph, in order to find out the

effectiveness and efficiency of privacy preserving data. Recently Privacy preserving data have acquired a great extent by data mining and Information security community researchers in development of techniques that integrates privacy concerns [4,5].

In recent years, number of cyber-attacks have been reported such as Denial of service, Trojan attacks, Man in the middle attack (MITM), Phishing attacks, Drive-by-downloads, hacking etc [6,7]. These attacks are implemented by use of Internet and so commonly known as Cybercrime. The increased number of attacks day by day is mainly due to broad use of social media websites, e-commerce websites and different applications people mostly spending time on it. Some of the major hacking tools to get unauthorized access to computer systems are Network Mapper (NMAP), Metasploit, Angry IP Scanner, Cain and Abel, Ettercap, Super scan etc.

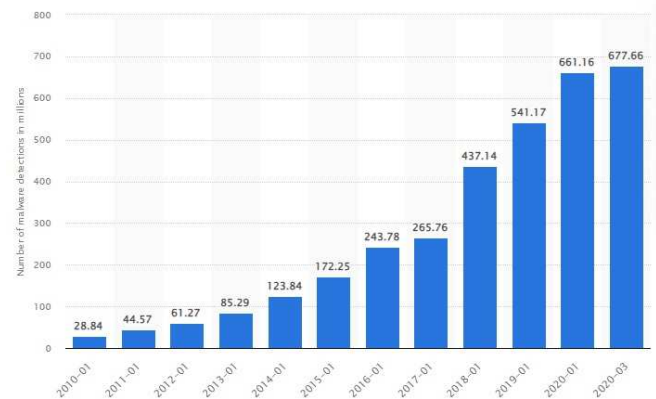


Fig 1. Total Malware infection Growth Rate (In Millions)

There is an urgent need of reinforcing the security measures of an organization by identifying the possible vulnerabilities at the level of network and host end. This requires the defenders to detect cyber threats and develop remedy to tackle the security issues. Attackers need to employ one vulnerable action to break the network security due to which defenders have to protect the system as a whole [8]. An Advanced Persistent Threat (APT) attacker makes use of malicious software programs to deploy attack in victim's system such as commands defined in Windows PowerShell can execute commands in remote systems, use of spear phishing E-mails, malicious websites, malicious advertising and Drive-By-Downloads are such examples [9]. In view of Cybersecurity most of the organizations shares threat information in form of reports.

- The first step in Cybersecurity is threat detection or finding malicious activities taken up by adversaries to target a victim's network,
 - Second step is to build software programs and to defend through responses, [10-12]
 - Last step is to take preventive measures and mitigate the risks associated with Cyber-attacks.
- Security programs should be adequate to analyze attack patterns rapidly and efficiently to take disciplinary actions quickly and so reduce comprising of the network and data. Conducting threat hunt is a common practice in Security Information and Event Management (SIEM), UEBA, ALERTING, and APT Prevention techniques [13,14].

Table 1: Red Teams vs Blue Teams

Analysis	Red Teams	Blue Teams
Working	Red teams perform penetration testing to exploit the vulnerabilities, uncovering all possible attack techniques with good intention to gain attackers insights in breaching the security controls	Blue teams work on defending the system from attacks by protecting the asset data against any threat activities.
Use of MITRE ATT&CK	MITRE ATT&CK matrix is a repository of tactics and techniques adopted by red teams to perform penetration testing deployed by ATP attackers.	Blue teams can strengthen the defense system by emulating attacker behaviors and mitigation approaches available in the repository.
Example	For an instance, a red team can use this framework to detect Credential Access threat by implementing techniques such as keyloggers or credential dumping.	A Blue team can address this attack by emulating mitigation techniques to protect from credential access attack such as Boot or Logon AutoStart Execution, OS Credential Dumping etc.

Cyber Threat Intelligence (CTI) is collecting the threat related information, processing and exploiting, analyzing the threat data with red teams and blue teams for decision making. MITRE proposed ATT&CK (Adversarial Tactics, Techniques and Common Knowledge) framework which defines adversarial behaviors and ways in which adversaries perform attacks based on real observations [15]. ATT&CK is a knowledgebase represented in matrix form which can be used to detect and response from threats. Table 1 shows the comparison of red teams and blue teams on the basis of working, use of MITRE ATT&CK framework and an example [16,17].

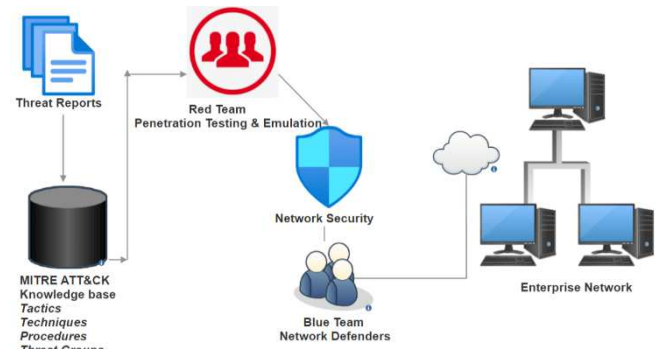


Fig 2. Threat detection and Response via Red Teams and Blue Teams in an enterprise network

Fig 2 shows how MITRE ATT&CK database gets updated with the tactics, techniques and procedures observed in threat reports and then red team basically uses this information to detect and emulate the threats performed by adversaries along with blue team which then focusses on the mitigation procedures to defend the cyberattacks and mitigate the risks involved on an enterprise network [18]. The paper is organized as follows. Section 2 discusses the lifecycle of a cyberattack. MITRE ATT&CK framework is discussed in Section 3. Section 4 defines adversarial simulation and ATT&CK. Section 5 discusses adversary emulation techniques and finally Section 6 concludes the paper [19,20].

II. THE LIFECYCLE OF AN ATTACK

The Cyberattack lifecycle also known as Cyber Kill-chain is defined by the following phases- Reconnaissance, Weaponization & Delivery, Exploitation, Installation, Command and Control, and finally Actions on the Objective. Fig 3 depicts the lifecycle of an attack.

1. **Reconnaissance**- In the first phase of Cyber-attack life cycle, the adversary plans for the attack, by scanning the target networks, identifying vulnerabilities and mission targets. Adversaries gain this information from different sources such as Facebook, LinkedIn, Twitter etc by capturing sensitive data from human's perspective [21,22].



Fig 3. The Lifecycle of an Attack.

2. **Weaponization & Delivery**- The next stage is weaponization where adversary is collecting the resources for implementing the attack. Attacker has to decide which methods should be used to deliver malicious packets over the network to infect the victim's system. Methods such as

spear phishing emails, malvertizing, exploit kits, etc. are commonly used weapons.

3. **Exploitation-** The third stage is where the attackers are deploying Cyberattacks using different weapons as discussed in the second stage and exploiting the victim's vulnerable system. This makes the adversary as an initial point of entry in the enterprise network [23,24].
4. **Installation-** As the attacker has entered in the network, next step is installation of malicious programs in defender's system to keep maintained control of access, persistence even if defender changes password the attacker can continue keep tracking of system, and escalating privileges to get highest privilege of system as administrator [25].
5. **Command and Control-** In the next stage of lifecycle, adversary can access remotely the infected system via commands and can control sending and receiving of data information between the attacker's malware network and the infected victim's network readily [26].
6. **Actions on the Objective-** Finally, As the attacker has full access control, persistence and escalating privileges now he can target in achieving objectives such as stealing of data, destruction of network infrastructure, creating fear etc [27].

III. MITRE'S ATT&CK FRAMEWORK

MITRE ATT&CK framework defines taxonomy for the defenders to build security programs for threat detection and response. Adversaries adopt different techniques to perform attacks such as penetration into the network, phishing activities, malware software links, data stealing, mal-advertising etc. to destroy the defender's network [28,29]. ATT&CK framework maps adversarial behaviors in to a structured matrix representation of tactics and techniques followed with procedures. Different Threat Groups such as APT3, APT19, APT 29 etc. have analyzed the adversarial activities and stated in the ATT&CK repository. Cyber Threat Intelligence (CTI) teams are using ATT&CK to analyze single adversary Group and identifying behaviors helps organizations to detect and defend from the attacks [30-33].

Table 2: Some Tactics and Techniques in MITRE ATT&CK Framework

Initial Access	Execution	Persistence	Privilege Escalation
9 techniques	12 techniques	19 techniques	13 techniques
Drive-by Compromise	Command and Scripting Interpreter	Account Manipulation	Abuse Elevation Control Mechanism
Exploit Public-Facing Application	Container Administration Command	BITS Jobs	Access Token Manipulation

External Remote Services	Deploy Container	Boot or Logon Autostart Execution	Boot or Logon Autostart Execution
Hardware Additions	Exploitation for Client Execution	Boot or Logon Initialization Scripts	Boot or Logon Initialization Scripts
Phishing	Inter-Process Communication	Browser Extensions	Create or Modify System Process
Replication Through Removable Media	Native API	Compromise Client Software Binary	Domain Policy Modification
Supply Chain Compromise	Scheduled Task/Job	Create Account	Escape to Host
Trusted Relationship	Shared Modules	Create or Modify System Process	Event Triggered Execution
Valid Accounts	Software Deployment Tools	Event Triggered Execution	Exploitation for Privilege Escalation
	System Services	External Remote Services	Hijack Execution Flow
	User Execution	Hijack Execution Flow	Process Injection
	Windows Management Instrumentation	Implant Internal Image	Scheduled Task/Job
		Modify Authentication Process	Valid Accounts
		Office Application Startup	
		Pre-OS Boot	
		Scheduled Task/Job	
		Server Software Component	
		Traffic Signaling	
		Valid Accounts	

In Table 2, a small tactics and techniques set can be visualized in MITRE ATT&CK framework. Figure 5 shows in general, there are 14 Enterprise tactics defined along with their description. Each tactic is implemented by means of number of techniques [34-36]. Enterprise matrix defines 185 techniques such as T1548, T1134 etc. in total and 367 sub-techniques such as T1548.001, T1548.002, T1548.003, T1548.004 etc. are the procedures to deploy these techniques in real world. There are 42 mitigations approaches defined which explains the prevention

measures to be taken when an adversary attack is attempted. For an instance, some of the techniques addressed by mitigation are T1110 Brute Force, .001 Password Guessing, .003 Password Spraying .004 Credential stuffing etc [37,38].

A. Tactics

A tactic is defined as a column header name in MITRE ATT&CK framework which specifies what adversary wants to perform an action. For example, Initial Access is a Tactic what adversary wants to gain access in the victim's network. For this, adversary can use different techniques such as spear phishing by sending fake Email messages with malicious links to the target to access their personal information in order to gain access in the network [39]. Each tactic is associated with different techniques and some techniques can be commonly deployed for achieving multiple tactics at the same time. For example, Abuse Elevation Control Mechanism is a technique which is common in Privilege Escalation and Defense Evasion tactics [40-42].

Table 3: ATT&CK tactics: Enterprise

ID	Name	Description
TA0043	Reconnaissance	The adversary is trying to gather information they can use to plan future operations.
TA0042	Resource Development	The adversary is trying to establish resources they can use to support operations.
TA0001	Initial Access	The adversary is trying to get into your network.
TA0002	Execution	The adversary is trying to run malicious code.
TA0003	Persistence	The adversary is trying to maintain their foothold.
TA0004	Privilege Escalation	The adversary is trying to gain higher-level permissions.
TA0005	Defense Evasion	The adversary is trying to avoid being detected.
TA0006	Credential Access	The adversary is trying to steal account names and passwords.
TA0007	Discovery	The adversary is trying to figure out your environment.
TA0008	Lateral Movement	The adversary is trying to move through your environment.
TA0009	Collection	The adversary is trying to gather data of interest to their goal.
TA0011	Command and Control	The adversary is trying to communicate with

		compromised systems to control them.
TA0010	Exfiltration	The adversary is trying to steal data.
TA0040	Impact	The adversary is trying to manipulate, interrupt, or destroy your systems and data.

B. Techniques

A technique is a method employed by an attacker to exhibit a particular threat activity in victim's network. ATT&CK gives many information about technique such as description, Examples, approaches for mitigation and detection, references. For example, one of the techniques for stealing user's account name and password that is, for getting Credential Access tactic implementation, Password Cracking technique, Password Spraying technique, Brute force techniques etc. can be used [43].

Fig 4. shows how an attacker may deploy one or more consecutive tactics using different techniques in order to achieve an objective. For example, if an attacker goal is to enter in to victim's network using spear phishing techniques, then to gain higher privileges using Process Injection techniques, then to gain personal account information using credential dumping techniques, and setting malicious software to the network using Scheduled Task, and finally to move laterally across the network using Pass the Hash techniques [44]. This includes predominantly the steps defined in ATT&CK tactics in sequence such as Initial Access, Escalating Privileges, Credential Access, Persistence and Lateral Movement [45].

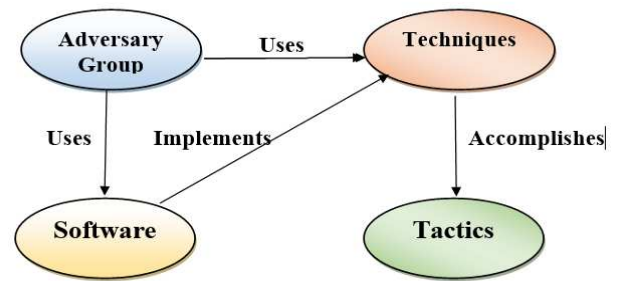


Fig 4. Threat Hunting using MITRE ATT&CK Framework

C. Procedures

A procedure is a step-by-step process for implementing Cyber threat attack techniques. An adversary uses different procedures to simulate an attack technique with software. For example, adversaries may follow different procedures to dump credentials to get user account details, in form of a hash or clear text password. These credentials can then be used for lateral movement across the victim's network. Table 4. shows an example of OS Credential Dumping Sub-Techniques.

Table 4: OS Credential Dumping Sub-Techniques.

ID	Name
T1003.001	LSASS Memory
T1003.002	Security Account Manager
T1003.003	NTDS
T1003.004	LSA Secrets
T1003.005	Cached Domain Credentials
T1003.006	DCSync
T1003.007	Proc Filesystem
T1003.008	/etc/passwd and /etc/shadow

Table 5. APT3 +APT29 techniques in ATT&CK Navigator

Reconnaissance	Resource Development	Initial Access	Execution	Persistence	Privilege Escalation	Defense Evasion
10 techniques	7 techniques	9 techniques	12 techniques	19 techniques	13 techniques	42 Techniques
Credential Access	Discovery	Lateral Movement	Collection	Command and Control	Exfiltration	Impact
16 techniques	30 techniques	9 techniques	17 techniques	16 techniques	9 techniques	13 Techniques

D. Comparing Layers in ATT&CK Navigator

ATT&CK navigator can be used to prioritize the techniques by comparing the techniques used by two different threat Groups in order to gather requirements by defenders to give to the CTI team for detailed investigation in detection purpose. If an organization consider certain Groups to be highly threat oriented for them, they can do defensive coverage analysis by prioritizing commonly used techniques through ATT&CK navigator. For example, if Real time threat techniques collected by different Groups such as APT3 and APT 29 are considered highly threat specific by an organization, these techniques can be mapped in navigator with specific colors.

To explore ATT&CK Navigator, first Creating a Layer named as APT3 and then selecting specific Threat Group from multi-select selection controls in the panel and give any score let us say 1 from technique controls. Second, create a new layer named APT29 and then selecting specific threat Group from multi-select selection controls in the panel and give score let us say 2 from technique controls. Finally, create a layer from other layers and mention score expression as APT3+APT29.

Fig 5. shows APT3 + APT29 techniques mapped in ATT&CK navigator in which red color highlights techniques used only in APT3, yellow color specifies the one used only by APT29 and green color visualizes common techniques in both of them. The defenders can track these commonly used techniques by adversaries and can clearly state this information to CTI team to prioritize resources for defense from these techniques.

IV. ADVERSARIAL SIMULATION AND ATT&CK

MITRE ATT&CK is useful to develop threat detection controls and prevention measures for each technique in the enterprise matrix. There are multiple ways to deploy an ATT&CK technique so if we block a single way there can be other ways feasible to perform this technique. One needs to employ techniques which can properly block entire threat coverage of an organization. For an instance, if anti-virus software is able to detect Mimikatz this doesn't mean it can cover from other attacks too such as Pass the Hash and Pass the Ticket, as there are other ways also to perform these techniques without involving the use of Mimikatz.

ATT&CK keep information of nearly 122 Threat Groups or Reports for detection and analysis of intrusion activities for Cyber threat Intelligence. Various organizations and analysts often record malicious attempts from adversaries in the form of Threat Groups or Reports and keeps complete information of publicly reported target domains, attack types, techniques employed to perform such attacks. Examples include APT1, APT12, APT29 etc. With this Threat Group information, organization can come to know what steps should be taken to defend and prevent from such threat hunting. In order to test the adversarial simulation techniques, the best way is to test security controls and their effective implementation, ensure full coverage against different techniques, check gaps in the existing protection scheme, check system configurations, live demonstration to catch where actors will succeed, etc.

Additionally, there are certain commercial as well as open-source tools available to simulate the attack techniques listed in enterprise matrix. Commercial tools like Verodin, SafeBreach, and AttackIQ are available to test adversarial activities internal to the environment aligned by ATT&CK. Open-source automated adversarial simulation tools such as UberMetta, MITRE Caldera, Red Canary Atomic Red Team, and Endgame Red Team Automation are available to simulate, hunt and detect the threats. All such tools are built on MITRE framework and assists Red team for automated adversary simulation.

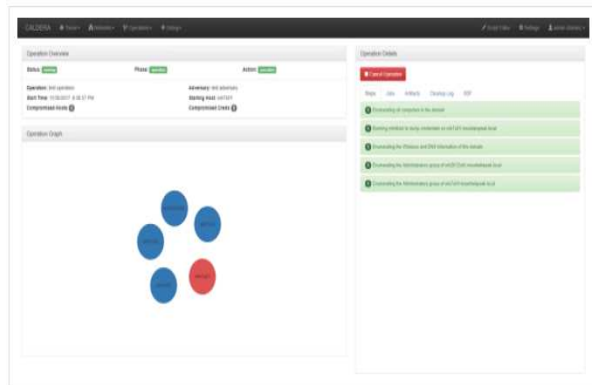


Fig 6. MITRE Caldera framework for adversary simulation

A. Identifying Data Sources

The collection, analysis and sharing of person specific data raises serious concerns about the privacy of individuals. The threat detection process begins with the availability of data sources. A data source is the source of information collected by a logging system to identify the action being performed, sequence of actions/ techniques performed by an adversary. This information allows red teams and blue teams to understand and validate the detection and prevention of specific adversarial techniques respectively. MITRE ATT&CK repository defines data sources requirements listed in each technique's procedures. For an instance, a technique named Credentials in Registry T1214 under Credential Access tactic lists different data sources such as Windows Registry, Process- Command-Line parameters, and Process monitoring as shown in Fig 7. All ATT&CK dataset is available in Structured Threat Information Expression (STIX) and can be accessed from Github repository or via public Trusted Automated Exchange of Intelligence Information (TAXII) server. This gives access to Python libraries such as cti-python-stix2 and cti-taxii-client developed by MITRE.

Credentials in Registry

The Windows Registry stores configuration information that can be used by the system or other programs. Adversaries may query the Registry looking for credentials and passwords that have been stored for use by other programs or services. Sometimes these credentials are used for automatic logons.

Example commands to find Registry keys related to password information:^[1]

- Local Machine Hive: `reg query HKLM /f password /s: REG_SZ /s`
- Current User Hive: `reg query HKCU /f password /t: REG_SZ /s`

ID: T1214
Tactic: Credential Access
Platform: Windows
Permissions Required: User, Administrator
Data Sources: Windows Registry, Process command-line parameters, Process monitoring

Fig 7. Data sources for T1214

B. Prioritizing of ATT&CK Data Sources

One can prioritize which MITRE techniques should be detected and simulated first as compared to other techniques by identifying data sources common for most of the techniques in a given environment. System for analyzing large volumes of log data and also provides as user interface to interpreted raw data analysis in an efficient way and to prevent abnormal behavioral attacks, sending alerts, time-based schedulers, and other common types of cyber-attacks. The top 10 data sources defined in Enterprise matrix techniques are named as Process

monitoring, File monitoring, Process command line parameters, API monitoring, Process use of network, packet capture, Windows Registry, Authentication logs, net flow, and Windows Event logs are often collected from Sysmon, Windows Event Logs, and other EDR platforms. Fig 8. shows a statistic of nearly 157 MITRE ATT&CK techniques used Process monitoring as a data source, around 90 techniques used File monitoring as a data source, and 87 techniques approximately used Process Command line parameters as a data source. If we implement ATT&CK techniques around these three data sources, detections can be maximized in a given environment.

ATT & CK Techniques with Data Sources

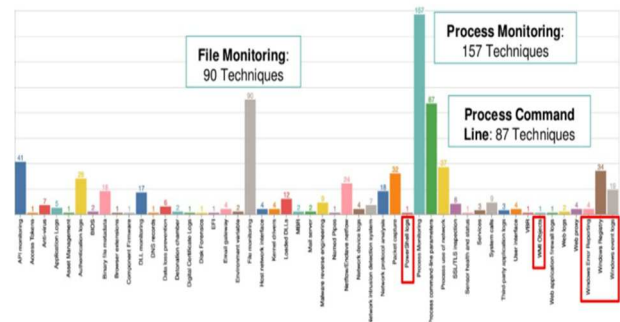


Fig 8. MITRE ATT&CK techniques with Data Sources

Fig 9. visualizes the count of techniques in MITRE framework versus data sources associated. Almost 200 techniques require two data sources to analyse them whereas only 19 (9%) techniques require one data source to analyse them.

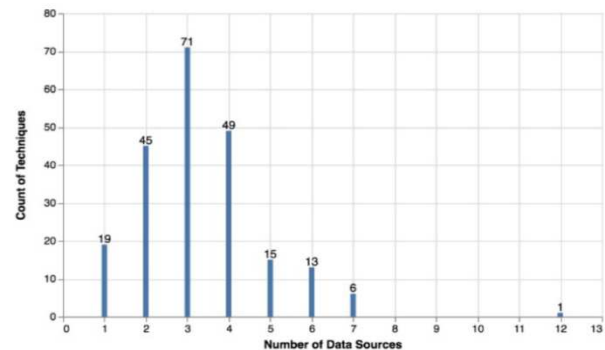


Fig 9. Data Sources associated with Count of techniques

All data sources listed in ATT&CK techniques refers to sub- data sources as well. Selection of data sources depends on which specific techniques are relevant to an organization in an environment and sub-data sources can be identified accordingly. Number of sub -data sources utilized in implementing the same ATT&CK technique may vary unconditionally. For an instance, an ATT&CK technique named Credentials in Registry T1214 under Credential Access Tactic maps to the data sources such as Windows Registry, process monitoring and process command-line parameters may have access to the various sub- data

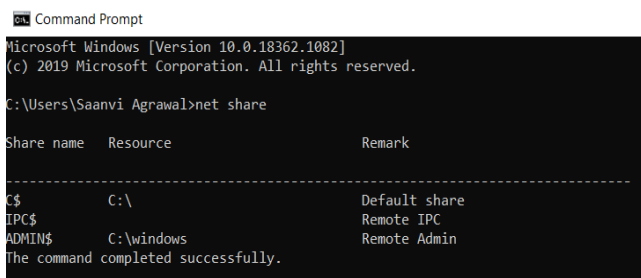
sources such as process creation, process termination, process write- to process, process access etc. Once we have data identified for analysis, we can import those data to run on some kind of search platform such as Security Information and Event Management (SIEM) to perform analytics and data simulation against them. After getting basic search returning data as results, next step is to reduce false positives in our environment. The goal should not be to get zero false positives instead to minimize them to get exact malicious behaviour.

V. ADVERSARY EMULATION

An adversary emulation is a step in cyber threat intelligence carried out by red teams that imitate possible threats to an organization in order to acquire behaviours and actions taken by adversary. In general, adversary emulators create a scenario to test malicious behaviours and various aspects of MITRE's tactics, techniques and procedures (TTPs). The red team observes this scenario to test for a target network and finds how defenses are fully protecting against the adversary emulation. This is how emulation techniques work individually from penetration testing taken up by red teaming. ATT&CK is a huge data repository about tactics techniques and procedures so it is convenient to analyse adversary and red team behaviours efficiently. Defense teams uses MITRE's framework for threat emulation to protect their organizations. Atomic Red team consists of small scripts or atomic test suites to simulate and test adversary behaviour to detect ATT&CK techniques and procedures across different platforms such as Windows, MAC OS, Linux etc. For an instance, a Chinese threat group named APT3 uses technique such as Network Share Discovery T1135 to identify shared files and folders on remote systems and also information required for lateral movement. Net should be used to query remote systems to know about shared files and folders specifically in the given format for Windows platform-

Attack Command- NET VIEW [\computer name [/CACHE] | [/ALL] | /DOMAIN [domain name]].

One can also use net command to identify shared drives on local system using net share as shown in Fig. 10.



```

Microsoft Windows [Version 10.0.18362.1082]
(c) 2019 Microsoft Corporation. All rights reserved.

C:\Users\Saanvi Agrawal>net share

Share name      Resource                Remark
-----
C$              C:\                    Default share
IPC$            C:\IPC                 Remote IPC
ADMIN$          C:\windows             Remote Admin
The command completed successfully.

```

Fig 10. Accessing shared drives in local system

Atomic Red team is used to detect adversary techniques and procedures and test behaviour analytic as expected to take preventive actions as and when required. When we execute atomic test named net view command to find if detections are made as expected is what we actually detected. Moreover, SIEM tool being used has behaviour analytic feature to check whether we arrive at the correct

solution what net view has flagged result, if any variation and SIEM tool didn't responded so we conclude that logs have not exported correctly from our host. Fix the problem and make desired improvements to know and catch the adversary closely using this procedure in future. These singular atomic tests detect adversary behaviour corresponding to an individual ATT&CK technique and procedure, allows red team to integrate with other techniques behaviour testing of overall adversary actions.

Fig 11. shows a chained process for atomic red team to begin adversary emulation for single technique testing and expands to other techniques for behaviour analytics. First step in this process is to create a scenario (commonly called abilities in Caldera) based on adversary attacks and choose a specific ATT&CK technique and its procedure to implement and synthesize behaviour of that attack.



Fig 11. Atomic testing cycle with ATT&CK

Caldera, an automated adversary emulation system consists of many built-in behaviours corresponding to ATT&CK techniques. This tool is useful to choose a technique or set of techniques when building the test and simulating behaviours of all coverage of attacks. Next step is to select a test for that technique based on which environment or platform the test will be executed say Linux, Windows or MAC OS. Now the test command will be executed and behaviour analysis can be done on any open source SIEM tool. This analysis leads us to detect malicious adversary actions with respect to the selected technique and procedures. Finally, this test results can be mapped with behaviour analytics defined in SIEM tools to make improvements if possible and later this cycle can be repeatedly performed on different ATT&CK techniques integrated to simulate overall behaviour of adversary actions.

VI. CONCLUSION AND FUTURE WORK

In this work, we investigated various cyber security issues by classifying real time threat reports in form of tactics, techniques and procedures used to defend from fastest growing cyber threats across the enterprises. The ATT&CK knowledge base is used as a fundamental principle for solving Advanced Persistent Threats (APT) problems by developing threat models and methodologies used in public, private sectors, and Cyber Threat detection and control communities to detect and mitigate the risk factors. To complete this work, we first discussed lifecycle of an attack, followed by in depth analysis of MITRE framework in defining TTPs and comparing various layers in ATT&CK navigator to map common techniques used by adversaries and use this information to prioritize our

resources how to defend against them. Second, adversary behavior simulation with ATT&CK begins by identifying data sources and prioritizing them to run on any SIEM tool to perform behavior analytics. Lastly, we showed how ATT&CK is used for red teaming and adversary emulation to analyze overall behavior of adversary actions. Our current work focusses on methods of threat detection and emulation techniques with MITRE framework. To extend our work in future, we will be investigating an approach to implement any of MITRE's techniques and procedures for attack detection and also ways to prevent and mitigate from such attacks.

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